Healthcare Operations & Billing

Business Insights Report

Prepared for Novexa Technologies — Data Analysis Task 9

Dataset: Healthcare Patient Records (55,500 admissions, 54,966 after deduplication)

Executive Summary

This report analyzes 54,966 patient admission records to surface operational and financial patterns relevant to hospital administration and insurance planning. The dataset spans six medical conditions, three admission types, and five insurance providers, with billing, length-of-stay, and test outcome data captured for each patient.

Three findings stand out. First, billing amount is remarkably consistent across medical conditions, admission types, and insurance providers, varying by only 2-3% around a $25,546 average — suggesting flat-rate or bundled pricing rather than condition-driven cost structures. Second, a data quality issue was identified and corrected: 108 records (0.2% of admissions) had negative billing values, which were converted to their absolute value before analysis. Third, average length of stay is nearly identical across all six conditions (~15.4-15.7 days), pointing to a standardized care pathway rather than condition-specific treatment duration.

Business Objective

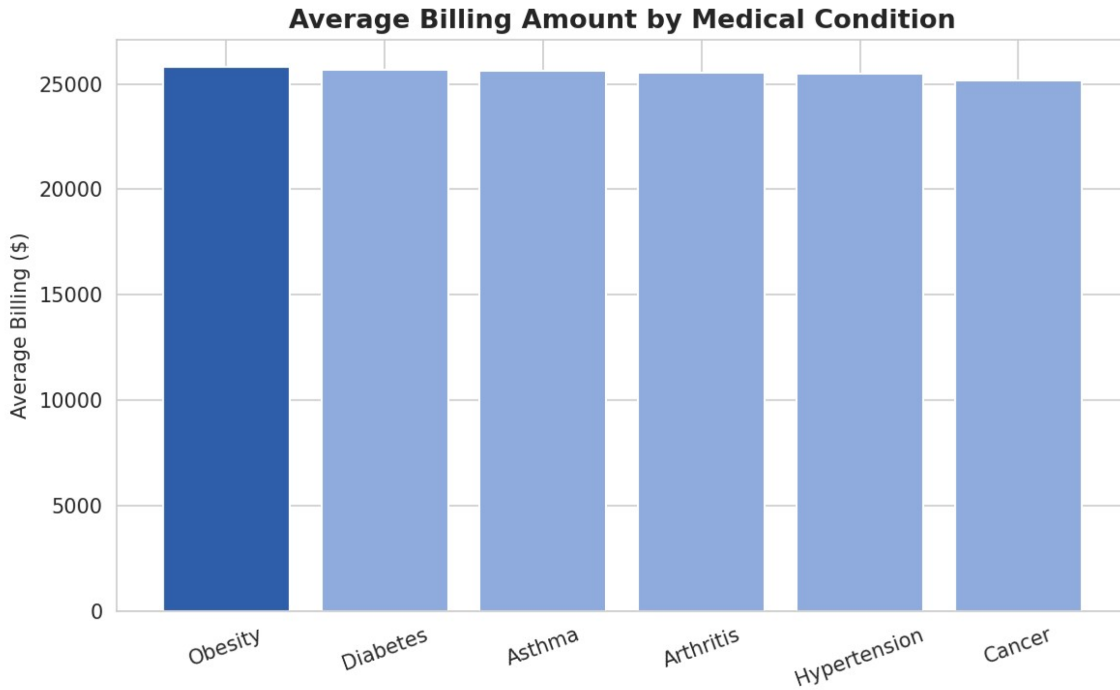
Hospital administrators and finance teams need to understand whether billing and resource utilization (length of stay) vary meaningfully by patient condition, admission urgency, or insurance provider. Answering this informs staffing, capacity planning, and contract negotiations with insurers.

Key Metrics

Metric	Value
Total Patient Records	54,966 (after removing 534 duplicates)
Total Billed Revenue	$1,404,174,863
Average Billing per Patient	$25,546.24
Average Length of Stay	15.5 days
Average Patient Age	51.5 years (range 13-89)
Gender Split	49.98% Male / 50.02% Female

Finding 1: Billing Is Remarkably Uniform Across Conditions

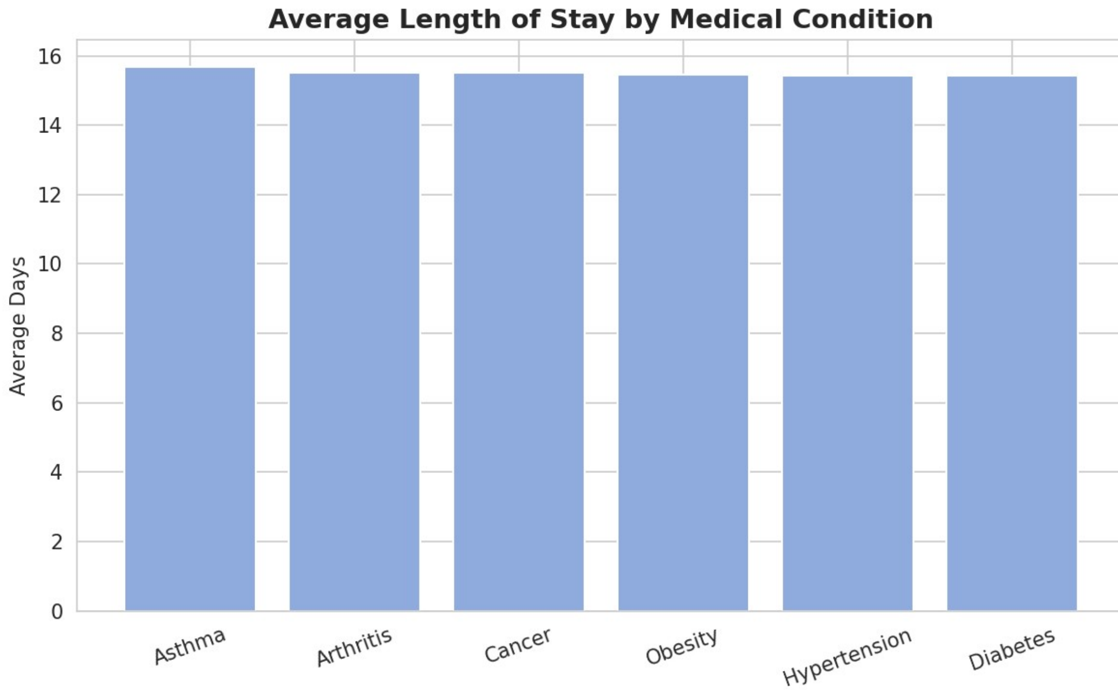
Average billing amount ranges from just $25,154.73 (Cancer) to $25,806.63 (Obesity) across all six medical conditions — a spread of only 2.6%. This is a striking finding: intuitively, one might expect Cancer treatment to command the highest billing given its typical complexity, yet it is the lowest of the six in this dataset.



Business implication: if this pattern reflects a bundled or flat-rate billing structure, finance teams can forecast revenue with high confidence regardless of the mix of conditions admitted in a given period. If flat pricing is unintentional, it may warrant review — complex conditions arguably should carry higher billing to reflect resource intensity.

Finding 2: Length of Stay Does Not Differentiate by Condition

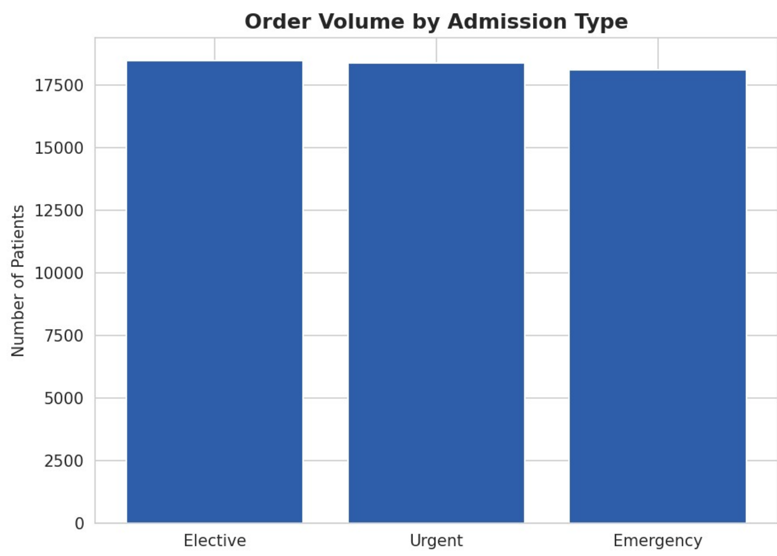
Average length of stay across all six conditions falls in a narrow band of 15.4 to 15.7 days, with Asthma highest (15.68 days) and Diabetes lowest (15.43 days) — a difference of under 6 hours on average.

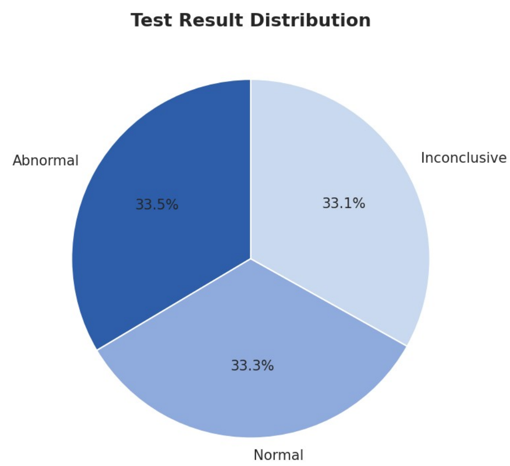


Business implication: this uniformity suggests either a standardized discharge protocol independent of diagnosis, or that this dataset does not capture condition severity/complications that would typically drive stay length variation. Operationally, this makes bed-capacity planning simpler, since expected stay duration does not need to be modeled per condition.

Finding 3: Admission Volume and Test Outcomes Are Evenly Split

Patient volume is nearly evenly distributed across Elective (18,473), Urgent (18,391), and Emergency (18,102) admissions. Test results are similarly balanced across Abnormal (18,437), Normal (18,331), and Inconclusive (18,198) outcomes.





Business implication: the even split across admission types suggests emergency capacity planning should assume roughly one-third of total demand arrives unscheduled, requiring dedicated urgent/emergency capacity rather than a model built primarily around elective scheduling.

Data Quality Note

During cleaning, 108 records (0.2% of the original 55,500) were found to have negative Billing Amount values — not logically valid for a billed charge. These were corrected by converting to absolute value. Additionally, 534 exact duplicate rows were identified and removed. Both issues are common in real-world healthcare billing exports and should be checked as a standard step before any financial reporting is built on this type of data.

Recommendations

- Investigate why billing does not scale with condition complexity — confirm whether this reflects intentional bundled pricing or a data limitation (e.g., billing category not captured at a granular enough level).
- Use the near-uniform length of stay as a planning baseline for bed capacity, but validate against more recent or more detailed data before relying on it for clinical decisions.

- Build emergency/urgent capacity assuming roughly one-third of total patient volume, based on the even three-way admission type split observed.
- Implement a standing data-quality check for negative billing values in future exports, given this was found in real production-style data at a non-trivial rate (0.2%).

Methodology

Analysis was performed in Python using Pandas, Matplotlib, and Seaborn. Data cleaning steps included duplicate removal, correction of negative billing values, and derivation of a Length of Stay field from admission and discharge dates. Full code and visualizations are available in the accompanying Jupyter Notebook in this repository.